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COMPARATIVE STUDY OF WEAR RESISTANCE AND SURFACE ROUGHNESS OF INJECTABLE VERSUS CONVENTIONAL COMPOSITE RESIN — IN VITRO STUDY

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DECLARATION

I declare that this thesis is my own work. It has not been submitted, in whole or in part, for any other degree or qualification. The experimental data are those of the laboratory study described herein. Sources used in the text are cited in the list of references.

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ABSTRACT

Background. Highly filled injectable composites are sold for load-bearing use. Their loss of mass and change in surface roughness after toothbrush abrasion still need a direct comparison with a conventional nanohybrid resin under the same laboratory conditions.

Aim. To compare percentage weight loss and arithmetic mean roughness (Ra) of Beautifil Flow Plus X F00, G-ænial Universal Injectable and Beautifil II LS after 10 000 toothbrushing cycles.

Methods. Thirty-six discs (10 mm diameter × 1 mm thick; n = 12) were brushed at 2 N in a Colgate Total slurry prepared after ISO guidance. Mass was recorded before and after abrasion. Ra was measured by contact profilometry. Percentage weight loss was compared by one-way ANOVA. Ra after brushing was compared by the Kruskal–Wallis test ($\alpha = 0.05$).

Results. Mean percentage weight loss was 1.49 ± 3.14 % for Beautifil Flow Plus X F00, 0.65 ± 0.37 % for G-ænial Universal Injectable and 0.41 ± 0.25 % for Beautifil II LS ($p = 0.329$). Mean Ra after brushing was 0.104 ± 0.017 µm, 0.100 ± 0.023 µm and 0.194 ± 0.050 µm respectively ($p < 0.001$). Each injectable resin was smoother than Beautifil II LS. The two injectables did not differ from each other. One Beautifil Flow Plus X F00 disc lost about 11.4 % of its mass and accounts for that group’s large standard deviation.

Conclusions. Under this toothbrushing protocol, gravimetric wear did not differ significantly among the three resins. Both injectable materials remained significantly smoother than the conventional nanohybrid. The injectable means stayed below the 0.2 µm plaque threshold. The nanohybrid mean sat on that line.

Keywords. Injectable composite; nanohybrid composite; toothbrush abrasion; wear; surface roughness; giomer.

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LIST OF ABBREVIATIONS

ANOVA Analysis of variance

ISO International Organization for Standardization

Ra Arithmetic mean roughness

SD Standard deviation

S-PRG Surface pre-reacted glass-ionomer

ΔRa Change in Ra (after − before)

CHAPTER 1

INTRODUCTION

Resin composites fail when the surface wears away or when what is left is rough. Wear flattens anatomy and can shorten the vertical dimension. Roughness holds plaque, takes stain and favours caries at the margin (1,2). Toothbrushing is a daily source of both. Dentifrice abrasive works as a third body. The resin-rich skin is removed first. Filler then stands proud or is pulled out. Mass falls. Ra rises (3,4).

Highly filled injectable resins were made so that a syringeable paste could be used in cavities that used to be reserved for packable composites. Filler load is raised, particle size is reduced, and silane treatment is improved until the paste still flows but no longer behaves like a classical low-filled flowable (5,6). G-ænial Universal Injectable (GC Corporation, Tokyo, Japan) and Beautifil Flow Plus X F00 (Shofu Inc., Kyoto, Japan) are sold on that claim. Beautifil II LS (Shofu Inc.) is a sculptable low-shrinkage nanohybrid giomer and is the conventional control in this work.

Independent data on these three products, tested together, under three-body toothbrush abrasion, are thin. Two-body chewing-simulator studies are not a substitute. They load a steatite antagonist at about 49 N. They measure volume, not toothbrush slurry (6). The present test is the other wear mode: ISO toothbrushing guidance, 2 N, 10 000 strokes, taken as about one year of twice-daily brushing (3,7,8).

An Ra of 0.2 µm is the usual threshold above which plaque retention rises (1). Surfaces smoother than that do not keep less plaque in a way that matters clinically. A restoration rougher than about 0.5 µm can be felt by the tongue (2). Those two numbers are the clinical frame for the roughness results.

1.1 Aim of the Study

To compare wear resistance, as percentage weight loss, and surface roughness (Ra) of two highly filled injectable composites with a conventional nanohybrid composite after standardised toothbrushing.

1.2 Objectives

1. To measure specimen mass before and after 10 000 brushing cycles and to calculate percentage and absolute weight loss.
2. To measure Ra before and after the same abrasion and to calculate ΔRa.
3. To compare the three materials with the statistical tests appropriate to each outcome.

1.3 Null Hypotheses

1. There is no significant difference in percentage weight loss among Beautifil Flow Plus X F00, G-ænial Universal Injectable and Beautifil II LS.
2. There is no significant difference in Ra after toothbrushing among the same three materials.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Wear of Restorative Resins

Wear is progressive loss of substance from two surfaces in relative motion (9). In the mouth the process is mixed. Attrition is two-body contact. Abrasion at non-contact sites is usually three-body, with food or toothpaste as the loose agent. Corrosion and fatigue run beside both (9,10). Laboratory ranking is useful. It is not a clinical lifetime.

Filler volume, size, shape, inter-particle spacing, the resin, the degree of conversion and the silane layer all matter (11,12). Condon and Ferracane varied those three factors in an oral-wear simulator. Wear rose once filler volume fell below 48 vol%. It rose in a straight line as the fraction of silane-treated filler was cut. Degree of cure moved wear in the expected direction; the step was smaller (12). Smaller particles and a tighter, well-coupled filler bed protect the matrix. Classical flowables, filled at about 40–55 wt%, wear faster than packable hybrids for that reason (5,13). Raising filler load and cutting particle size was the route to the newer injectables. G-ænial Universal Injectable is listed at 50 vol %. That sits just above Condon's 48 vol% step. Beautifil II LS is listed at 83 wt%.

Toothbrushing tests follow ISO/TR 14569-1. The brush load should lie between 0.5 N and 2.5 N. The slurry is made from a dentifrice that meets ISO 11609 (3,14). Ten thousand strokes are widely treated as one year if a patient makes about fifteen strokes a surface, twice a day (7,8). The conversion is an estimate. It is still the figure most toothbrushing papers use.

Wear may be read as volume, profile depth or mass. Gravimetric loss is simple and repeatable if the balance is fine enough and the specimen is dry. It does not describe the shape of the worn surface. Volume from a scan does. The two numbers answer different questions (10,15).

2.2 Surface Roughness

Ra is the mean absolute deviation of the profile from the centre line. It is the parameter almost every dental paper reports. Contact stylus profilometry remains a standard method. Optical methods add a map but are not required to rank Ra (4,16).

Bollen, Lambrechts and Quirynen set 0.2 µm as the plaque threshold (1). Quirynen and Bollen had already shown that roughness and surface-free energy govern plaque on restorations (17). Jones, Billington and Pearson later asked volunteers to rank composite discs with the tongue. Discrimination sat between 0.25 µm and 0.50 µm. They advised finishing to no more than 0.50 µm if the patient is not to feel the surface (2).

Toothbrushing raises Ra. Heintze and co-workers showed that the rise depends on material, time and load (4). Nanofilled resins, with small particles that wear with the matrix, usually stay smoother than nanohybrids, in which larger fillers are uncovered or lost (8,18). The toothpaste matters. A higher relative dentine abrasivity cuts gloss faster and leaves a rougher track (8).

2.3 Injectable Composite Resins

Early flowables were not intended for occlusal load (13). The current injectables are a different paste. G-ænial Universal Injectable is filled to about 69 wt% (50 vol%) with ultrafine barium glass and silica, mean size about 150 nm, in a UDMA / Bis-MEPP / TEGDMA matrix (6,18). Beautifil Flow Plus X F00 is a zero-flow giomer injectable with nano S-PRG filler (19). It is not the same product as Beautifil Flow Plus F00 used in several earlier papers (6).

Rajabi et al. compared G-ænial Universal Injectable and Beautifil Flow Plus F00 with a conventional flowable and a nanohybrid paste in two-body wear (200 000 cycles, 49 N, steatite). The two injectables lost less volume than the paste and the classical flowable. Flexural strength was also higher. The two injectables did not differ from each other (6). That is occlusal contact wear, dry, measured as volume. It cannot be read as toothbrush mass loss.

Hasan, Eltoukhy and Zaghloul brushed G-ænial Universal Injectable, a nanofilled paste and a nanohybrid. After times taken as six months and one year, the injectable had the lowest Ra. The nanohybrid was the roughest. The two nanofilled resins did not differ (18). That design is closer to the present study.

Tzimas and co-workers reviewed highly filled flowables used as sole restoratives. Ra values were often statistically comparable to nanohybrid pastes, and sometimes lower, depending on filler packing and polish (20). Material-to-material scatter remains the rule.

2.4 Conventional Nanohybrid Resins and Giomers

Nanohybrid pastes mix nano-sized particles with larger submicron or micron fillers. Total inorganic load commonly exceeds 70–80 wt% (11). Beautifil II LS is such a paste. Filler load is 83 wt%. The resin uses an SRS monomer aimed at low shrinkage. The filler includes S-PRG glass (19). S-PRG releases ions, including fluoride. The same ion exchange can roughen the surface in acid (21). Larger and irregular fillers, once the matrix is brushed off, leave a higher Ra than a uniform nano-filled injectable (18). An eight-year clinical series of a giomer paste (Beautifil with a self-etch primer) showed acceptable retention and little secondary caries in the teeth that were still available for recall (22). That is clinical behaviour of an earlier giomer, not toothbrush Ra of Beautifil II LS. It only shows that S-PRG pastes have been used as definitive restoratives for years.

Packable nanohybrids remain the clinical reference for posterior direct work. They are harder to adapt in a narrow box and more likely to trap voids. The injectable products were written to close that handling gap without giving up wear resistance. Whether they do so under toothbrush abrasion is the question of this thesis.

2.5 Statement of the Problem

Two-body wear of G-ænial Universal Injectable and Beautifil Flow Plus F00 is favourable in one recent laboratory study (6). Toothbrush Ra of G-ænial Universal Injectable is favourable against a nanohybrid in another (18). Beautifil Flow Plus X F00 and Beautifil II LS have not been placed beside G-ænial Universal Injectable in the same 10 000-cycle, 2 N, gravimetric-plus-Ra protocol. That is the gap the experiment fills.

CHAPTER 3

MATERIALS AND METHODS

3.1 Study Design and Sample

The study was an in-vitro comparison of three resin composites. Thirty-six discs were prepared, 10 mm in diameter and 1 mm thick, in a CAD/CAM Teflon mould. The discs were allocated at random to three groups of twelve. A first G*Power estimate had given ten discs a group. Twelve were used to raise power.

3.2 Materials

The materials used were:

1. G-ænial Universal Injectable, a highly filled injectable composite (GC Corporation, Tokyo, Japan).
2. Beautifil Flow Plus X F00, a highly filled injectable giomer (Shofu Inc., Kyoto, Japan).
3. Beautifil II LS, a conventional nanohybrid giomer (Shofu Inc., Kyoto, Japan).
4. Colgate Total toothpaste.
5. Distilled water.

The composition of the three resin composites is summarised in Table 3.1.

Table 3.1 Brand, type, matrix, filler and load of the resin composites used in the study

Composite	Type	Matrix and filler	Load
G-ænial Universal Injectable	Nanofilled injectable	UDMA, Bis-MEPP, TEGDMA; ultrafine barium glass and silica (~150 nm)	69 wt% / 50 vol%
Beautifil Flow Plus X F00	Giomer injectable, zero flow	Dimethacrylate resin; nano S-PRG filler	Highly filled injectable
Beautifil II LS	Nanohybrid giomer	SRS monomer; S-PRG and pre-polymerised fillers	83 wt%

Manufacturers: GC Corporation, Tokyo, Japan; Shofu Inc., Kyoto, Japan.

3.3 Specimen Preparation

Each material was placed in the Teflon mould. A matrix strip and a glass plate were used to exclude air from the surface and to keep the disc flat. Polymerisation followed the manufacturer’s instructions for that product. Flash was removed. The discs were identified by group and stored in distilled water until testing.

3.4 Baseline Measurements

Each disc was dried and weighed on an analytical balance. The recorded masses run to 0.0001 g; the balance therefore read to 0.1 mg. Ra was measured with a contact profilometer on the test face. The value used for analysis was the mean of the traces taken on that disc.

3.5 Toothbrushing Protocol

Discs were mounted in a toothbrushing simulator. A slurry of Colgate Total and water was prepared after ISO 11609 and ISO/TR 14569-1 (3,14). The brush load was 2 N, inside the ISO range of 0.5–2.5 N (3). Each disc received 10 000 cycles. That count is the conventional laboratory equivalent of one year of twice-daily brushing (7,8). Fresh slurry was used as required to keep the abrasive present. After the run the discs were rinsed free of paste.

3.6 Post-test Evaluation

Mass and Ra were recorded again with the same instruments and the same drying habit as at baseline.

Percentage weight loss was calculated as:

$$\text{Percentage weight loss} = [(W_1 - W_2) / W_1] \times 100$$

where W_1 and W_2 are mass before and after brushing.

Absolute loss (mg) = $(W_1 - W_2) \times 1000$.

ΔRa (µm) = Ra after − Ra before.

3.7 Statistical Analysis

Descriptive values are presented as mean ± standard deviation. Percentage weight loss was compared among groups by one-way analysis of variance (ANOVA). Ra after brushing was compared by the Kruskal–Wallis test, followed by pairwise post-hoc comparisons. The level of significance was set at $\alpha = 0.05$.

CHAPTER 4

RESULTS

Thirty-six discs were tested, twelve in each group. All discs completed 10 000 cycles at 2 N. Wear is reported as mass. Roughness is reported as Ra.

4.1 Weight Loss

Mean mass before and after brushing, and percentage loss, are given in Table 4.1.

Table 4.1 Mean weight before and after simulated toothbrushing and percentage weight loss of the tested composite resins (n = 12)

Composite resin	**Weight before (g) Mean ± SD**	**Weight after (g) Mean ± SD**	**Weight loss (%) Mean ± SD**	**p-value**
Beautifil Flow Plus X F00	0.1602 ± 0.0074	0.1576 ± 0.0031	1.49 ± 3.14	
G-ænial Universal Injectable	0.1630 ± 0.0102	0.1619 ± 0.0106	0.65 ± 0.37	0.329†
Beautifil II LS	0.2313 ± 0.0154	0.2303 ± 0.0156	0.41 ± 0.25	

† One-way ANOVA. Values are mean ± standard deviation.

Beautifil II LS discs were heavier at baseline (0.2313 ± 0.0154 g) than the two injectables (0.1602 ± 0.0074 g and 0.1630 ± 0.0102 g). That follows from filler load and density, not from a difference in disc size.

One-way ANOVA on percentage loss gave $p = 0.329$. The first null hypothesis is not rejected. Numerically, Beautifil II LS lost the least (0.41 ± 0.25 %), then G-ænial Universal Injectable (0.65 ± 0.37 %), then Beautifil Flow Plus X F00 (1.49 ± 3.14 %). The last standard deviation is an order larger than the other two.

Absolute loss is given in Table 4.3, after the roughness data, because percentage loss on a lighter disc can exaggerate a small mass change.

4.2 Surface Roughness

Mean Ra before and after brushing, and ΔRa, are given in Table 4.2.

Table 4.2 Mean surface roughness (Ra) before and after simulated toothbrushing and the change in roughness (ΔRa) of the tested composite resins (n = 12)

Composite resin	Ra before (µm) Mean ± SD	Ra after (µm) Mean ± SD	ΔRa (µm) Mean ± SD	p-value
Beautifil Flow Plus X F00	0.038 ± 0.011	0.104 ± 0.017	0.066 ± 0.021	
G-ænial Universal Injectable	0.054 ± 0.010	0.100 ± 0.023	0.046 ± 0.023	< 0.001‡
Beautifil II LS	0.119 ± 0.054	0.194 ± 0.050	0.075 ± 0.066	

‡ Kruskal–Wallis test. Pairwise tests: each injectable versus Beautifil II LS, $p < 0.001$; the two injectables, not significant. Values are mean ± standard deviation.

At baseline Beautifil II LS was already the roughest (0.119 ± 0.054 µm). Beautifil Flow Plus X F00 was the smoothest (0.038 ± 0.011 µm). G-ænial Universal Injectable lay between them (0.054 ± 0.010 µm).

Every group roughened. ΔRa was 0.066 ± 0.021 µm, 0.046 ± 0.023 µm and 0.075 ± 0.066 µm. After 10 000 cycles the means were 0.104 ± 0.017 µm (Beautifil Flow Plus X F00), 0.100 ± 0.023 µm (G-ænial Universal Injectable) and 0.194 ± 0.050 µm (Beautifil II LS).

The Kruskal–Wallis test on Ra after brushing was significant ($p < 0.001$). The second null hypothesis is rejected. Pairwise tests separated Beautifil II LS from each injectable. They did not separate the two injectables.

Relative to the 0.2 µm plaque line (1), both injectable means were below it. The Beautifil II LS mean was 0.194 µm, on the line. Its standard deviation of 0.050 µm means some discs in that group crossed 0.2 µm. All three means were well below 0.50 µm (2).

Absolute weight loss is set out in Table 4.3.

Table 4.3 Absolute weight loss after 10 000 brushing cycles (n = 12)

Composite resin	Absolute weight loss (mg) Mean ± SD	Range (mg)
Beautifil Flow Plus X F00	2.58 ± 5.91	0.2 – 20.7
G-ænial Universal Injectable	1.07 ± 0.62	0.2 – 1.9
Beautifil II LS	0.95 ± 0.58	0.1 – 1.8

Values are mean ± standard deviation. The large standard deviation in Beautifil Flow Plus X F00 comes from one disc that lost about 11.4 % of its mass (20.7 mg). The rest of that group, and both other groups, sit in a band of roughly 0.1–1.9 mg.

4.3 Summary of Results

Percentage weight loss did not differ significantly. Absolute loss in the two well-behaved groups and in most of the Beautifil Flow Plus X F00 discs was about 1 mg. One disc in that group did not. After the same abrasion both injectables were smoother than Beautifil II LS and did not differ from each other.

CHAPTER 5

DISCUSSION

The first null hypothesis stands. The second does not. The three resins lost a statistically similar fraction of their mass. They did not finish with a similar Ra.

5.1 Weight Loss

A non-significant ANOVA ($p = 0.329$) is not proof of equal wear. It is a failure to reject equality. Beautifil Flow Plus X F00 had a standard deviation of 3.14 % beside a mean of 1.49 %. That spread is the outlier disc (about 11.4 %, 20.7 mg). Residual variance rises. Power falls. The other eleven discs in that group, and the twenty-four discs in the other groups, lost between 0.1 mg and 1.9 mg. On that restricted view the three pastes are close.

Beautifil II LS started heavier. Percentage loss on a heavier disc is a smaller number for the same milligrams shed. Absolute loss treats that fairly: 0.95 ± 0.58 mg against 1.07 ± 0.62 mg for G-ænial Universal Injectable. Those two means are near each other. The injectable with the nano-filled barium-glass bed did not shed more mass than the 83 wt% nanohybrid under this slurry.

Rajabi et al. found less volumetric two-body wear for G-ænial Universal Injectable and Beautifil Flow Plus F00 than for a nanohybrid paste (6). The direction is friendly to injectables. The test is not the same test. They used 200 000 chewing cycles, 49 N and a steatite ball. They did not use toothpaste. They measured volume. Beautifil Flow Plus F00 is also not Beautifil Flow Plus X F00. Agreement can be stated only as a general remark: highly filled injectables are not automatically the weaker partner in a wear assay.

Why one disc lost 20.7 mg is not known from the laboratory record used here. A void, a soft under-cured layer, or a weighing error would all produce that point. The value was not dropped. It is footnoted in Table 4.3. Any later re-analysis that excludes it must say so.

5.2 Surface Roughness

The roughness result is cleaner. The nanohybrid was rougher before brushing and rougher after it. ΔRa was also largest in that group, with a wide spread (0.075 ± 0.066 µm). G-ænial Universal Injectable changed the least (0.046 ± 0.023 µm). Beautifil Flow Plus X F00 started smoothest and ended statistically with G-ænial Universal Injectable.

That pattern fits filler geometry. A nanofilled injectable with ~150 nm barium glass and silica wears more evenly (6,18). A nanohybrid with larger S-PRG and pre-polymerised particles loses matrix first. The large fillers stand up or leave pits (4,18). Hasan, Eltoukhy and Zaghloul saw the same order after simulated brushing: G-ænial Universal Injectable smoother than a nanohybrid; two nanofilled resins not different from each other (18). Heintze et al. had already shown that hybrid and microhybrid pastes roughen more with load and time than microfilled resins (4).

The 0.2 µm line is the clinically useful one (1,17). Both injectables ended near 0.10 µm. They remain, on the mean, in the range where further smoothing is not expected to cut plaque. Beautifil II LS ended at 0.194 ± 0.050 µm. The mean is acceptable on the published threshold. The spread is not entirely below it. None of the means approach the 0.50 µm tongue threshold (2). Patients would not be expected to feel these surfaces after one year of this brushing model. Biofilm risk is the sharper distinction, and it applies to the nanohybrid group.

5.3 Clinical Implications

Within this protocol the two injectables may be used where toothbrush abrasion is the wear of interest without paying a roughness penalty against Beautifil II LS. They do not, on these data, wear less. They finish smoother. That matters on free cleanable surfaces and at margins. It does not license a claim that they survive occlusal contact better. Occlusal contact was not tested.

Beautifil II LS remains a reasonable packable control. Its higher baseline Ra may reflect filler size, polish, or both. If the nanohybrid was left with a coarser finish, part of the after-brushing difference was already present at baseline. The after-brushing test is still valid. The groups were compared as they stood after the same abrasion.

5.4 Strengths of the Study

The three products are current commercial pastes, including the X-generation Flow Plus. Sample size was twelve, not ten. Both mass and Ra were read on the same discs. The brushing load and cycle count sit on published ISO and Sexson–Phillips figures (3,7). The statistical tests match the distributions used in the locked spreadsheet. The outlier was kept and named.

5.5 Limitations of the Study

The test is in vitro. Saliva, pH, enzymes and occlusal load are missing. Gravimetric loss is not worn volume and is not worn anatomy. One dentifrice was used. There are no scanning electron images, so filler plucking is an inference from the roughness numbers, not a micrograph. Ten thousand cycles are one conventional year, not a clinical year. Disc geometry is not a Class II box.

5.6 Concluding Remarks

After 10 000 strokes at 2 N the three resins could not be separated on percentage weight loss. They could be separated on Ra. The injectables were smoother. Beautifil II LS sat on the plaque threshold. That is the result. It should not be enlarged into a general ranking of injectable versus nanohybrid outside toothbrush abrasion.

CHAPTER 6

CONCLUSIONS AND RECOMMENDATIONS

6.1 Conclusions

Under the conditions of this in-vitro study:

1. Percentage weight loss after 10 000 toothbrushing cycles did not differ significantly among Beautifil Flow Plus X F00, G-ænial Universal Injectable and Beautifil II LS ($p = 0.329$).

2. Ra after the same abrasion did differ ($p < 0.001$). Both injectable resins were smoother than Beautifil II LS. The two injectables did not differ from each other.

3. Mean Ra of both injectables remained below 0.2 µm. Mean Ra of Beautifil II LS was 0.194 µm.

4. One Beautifil Flow Plus X F00 disc lost about 11.4 % of its mass and is the source of that group’s large standard deviation.

5. The first null hypothesis is retained. The second is rejected.

6.2 Recommendations

Clinical practice. Where toothbrush abrasion and smoothness after hygiene are the concern, either injectable tested here may be considered beside Beautifil II LS. The data do not show less wear. They show a smoother surface after this brushing model.

Further laboratory work. Repeat the mass analysis with and without the outlying disc, and report both. Add volumetric loss or profilometric wear depth. Add SEM of the worn face. State the polish sequence, the brush and the slurry ratio in full. A second dentifrice, or a higher cycle count, would test whether the Ra gap holds.

Clinical research. Only a controlled clinical study can say whether the 0.10 µm versus 0.19 µm difference after a laboratory year changes plaque, stain or survival.

CHAPTER 7

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